Real and Complex Number Systems (Practice questions from Lecture 1, 2 and 3)

Section 1: Real Numbers and Decimals

1. Identify which of the following are rational and irrational numbers:
(i) $\sqrt{3}$
(ii) 1/6
(iii) π
(iv) 15/2
(v) 7.25
(vi) $\sqrt{29}$

2. Convert the following fractions into decimal fractions:
(i) 17/25
(ii) 19/4
(iii) 57/8
(iv) 205/18
(v) 5/8
(vi) 25/38

3. State whether the following statements are True or False:

(i) 2/3 is an irrational number.
(ii) π is an irrational number.
(iii) 1/9 is a terminating fraction.
(iv) 3/4 is a terminating fraction.
(v) 4/5 is a recurring fraction.

4. Represent the following numbers on the number line:
(i) $\frac{2}{3}$ (ii) $\frac{-4}{5}$ (iii) $1\frac{3}{4}$ (iv) $-2\frac{5}{8}$ (v) $2\frac{2}{3}$ (vi) $\sqrt{5}$

5. Give a rational number between 3/4 and 5/9.

6. Express the following recurring decimals as a rational number p/q:
(i) $0.\overline{5}$ (ii) $0.1\overline{3}$ (iii) $0.6\overline{7}$

Section 2: Radicals and Indices

7. Write each radical expression in exponential notation and vice versa:
(i) $\sqrt[3]{-164}$
(ii) $(2)^{3/5}$
(iii) $(-7)^{1/3}$
(iv) $(y)^{-2/3}$

8. State True or False:
(i) $5^{1/5} = \sqrt{5}$
(ii) $2^{2/3} = \sqrt[3]{4}$
(iii) $\sqrt{49} = \sqrt{7}$
(iv) $\sqrt[3]{x^{27}} = x^3$

9. Simplify:

(i) $\left(\frac{8}{125}\right)^{-4/3}$ (ii) $\frac{4(3)^n}{3^{n+1} - 3^n}$

10. Simplify:

(i) $\frac{(243)^{-2/3}\,(32)^{-1/5}}{\sqrt{(196)^{-1}}}$ (ii) $(2x^5\,y^{-4})\,(-8x^{-3}\,y^2)$

<u>Section 3: Complex Numbers</u>

11. Perform operations:
(i) (2 + 3i) + (7 - 2i)
(ii) 2(5 + 4i) - 3(7 + 4i)
(iii) -(-3 + 5i) - (4 + 9i)
(iv) $2i^2 + 6i^3 + 3i^{16} - 6i^{19} + 4i^{25}$

12. Simplify (a + bi form):

(i) $(-7 + 3i)\,(-3 + 2i)$ (ii) $(2 - \sqrt{-4})\,(3 - \sqrt{-4})$

(iii) $(\sqrt{5} - 3i)^2$ (iv) $(2 - 3i)\,(\overline{3 - 2i})$

13. Simplify (a + bi form):

(i) $\frac{-2}{1+i}$ (ii) $\frac{2+3i}{4-i}$ (iii) $\frac{9-7i}{3+i}$

(iv) $\frac{2-6i}{3+i} - \frac{4+i}{3+i}$ (v) $\left(\frac{1+i}{1-i}\right)^2$ (vi) $\frac{1}{(2+3i)\,(1-i)}$

14. Solve for x and y:

(i) $(2 - 3i)\,(x + yi) = 4 + i$
(ii) $(3 - 2i)\,(x + yi) = 2(x - 2yi) + 2i - 1$
(iii) $(3 + 4i)^2 - 2(x - yi) = x + yi$

Section 4: Challenge Sums

Q1. Simplify the following rational expressions to their lowest terms:

(i) $\dfrac{x-6}{x^2-36}$

(ii) $\dfrac{(x+4)^2}{(x+4)(x+7)}$

Q2. Expand and simplify the following products using algebraic identities:

(i) $(2x-18)(3x+3)$

(ii) $(4x-1)^2$

(iii) $(2x+7)(2x-7)$

(iv) $(3x-1)(2x+7y-9)$

Q3. Resolve the following expressions into factors completely:

(i) $81y^2-100$

(ii) x^3+512

(iii) $8x^3-125$

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difference of squares	$a^2-b^2=(a+b)(a-b)$
perfect square trinomial	$a^2+2ab+b^2=(a+b)^2$
sum of cubes	$a^3+b^3=(a+b)(a^2-ab+$
difference of cubes	$a^3-b^3=(a-b)(a^2+ab+$

Key Equations

Rules of Exponents

For nonzero real numbers a and b and integers m and n

Product rule	$a^m \cdot a^n = a^{m+n}$
Quotient rule	$\frac{a^m}{a^n} = a^{m-n}$
Power rule	$(a^m)^n = a^{m \cdot n}$
Zero exponent rule	$a^0 = 1$
Negative rule	$a^{-n} = \frac{1}{a^n}$
Power of a product rule	$(a \cdot b)^n = a^n \cdot b^n$
Power of a quotient rule	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

perfect square trinomial	$(x+a)^2=(x+a)(x+a)=x^2+2ax+a^2$